Methodological Paper Greece 1900-1950-2000

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INTRODUCTION

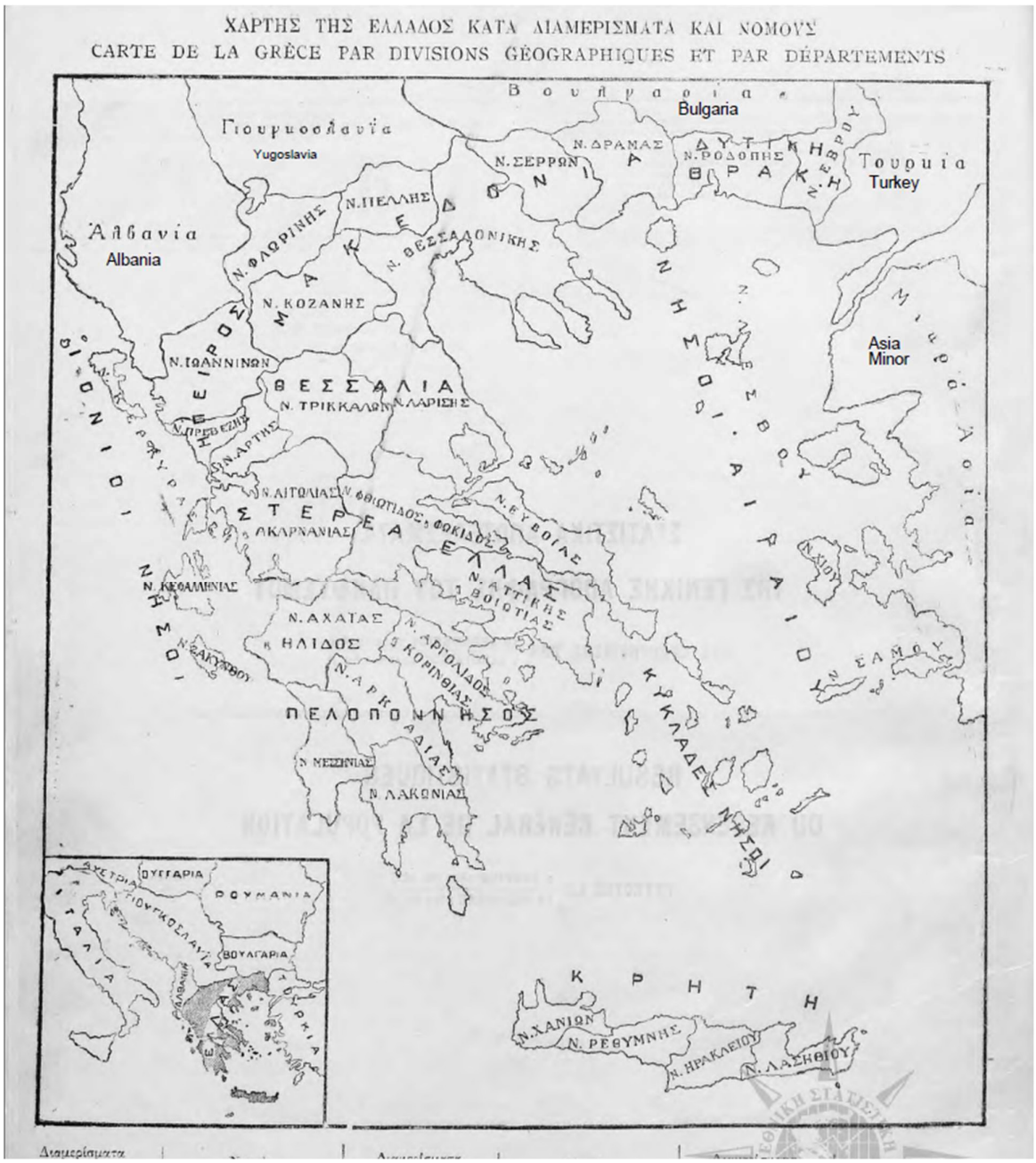
This paper aims to demonstrate the methodology and sources used to create the datasets for Greece's labor relations in the cross-sections 1900, 1950, and 2000. The data included in the datasets have been collected from the official state censuses as published by the Hellenic Statistical Authority (ELSTAT)[1] in the years 1920, 1928, 1951, 1991, and 2001. The 1920 population census was used for the cross-section 1900, the 1928 and 1951 for the cross-section 1950, and the 1991 and 2001 ones for the cross-section 2000. The choice of this particular categorization across the different cross-sections followed Greece's major sociopolitical events during those years. For instance, the 1922 Asia Minor Catastrophe and the refugee influx it caused were the most significant historical turning points to be considered. Thus, the 1920 census was accounted closer to the 1900 cross-section, and the 1928 census was deemed closer to the 1950 cross-section, due to the rapid population growth that occurred after 1922.

Greece has experienced multiple territorial alterations since its creation. The kingdom of Greece in 1832 included Peloponnesus, what is now regarded as Central Greece, and the Cyclades. In 1864, Great Britain ceded the Ionian Islands to Greece. Following, in 1881, the regions of Thessaly and Arta became part of the Greek state, while in 1913, the whole of Epirus and Macedonia, the North-eastern Aegean islands, and Crete were added. With the Treaty of Sèvres, after the end of World War I, Greece expanded eastward, annexing all of Thrace and the region of Smyrna. However, these regions soon became part of Turkey with the Treaty of Lausanne in 1923. Specifically, Smyrna (Izmir now), East Thrace, and the islands Imbros and Tenedos were ceded to Turkey. The data available for the 1920 cross-section, in this database, refer to the Greek state after the Lausanne Treaty. The course of Greece's territorial expansion is evident in the maps below.

[1] The first census used for this research (the 1920 population census) was carried out by the Central Division of Statistics, which was part of the Ministry of National Economy. In 1925, the General Statistical Service of Greece was established within the Ministry of National Economy (NSSG). The name ELSTAT comes from the establishment of the Hellenic Statistical Authority in 2010 as an independent authority.

MAPS

Map 1: The territory of the Greek State in 1923 after the Treaty of Lausanne.



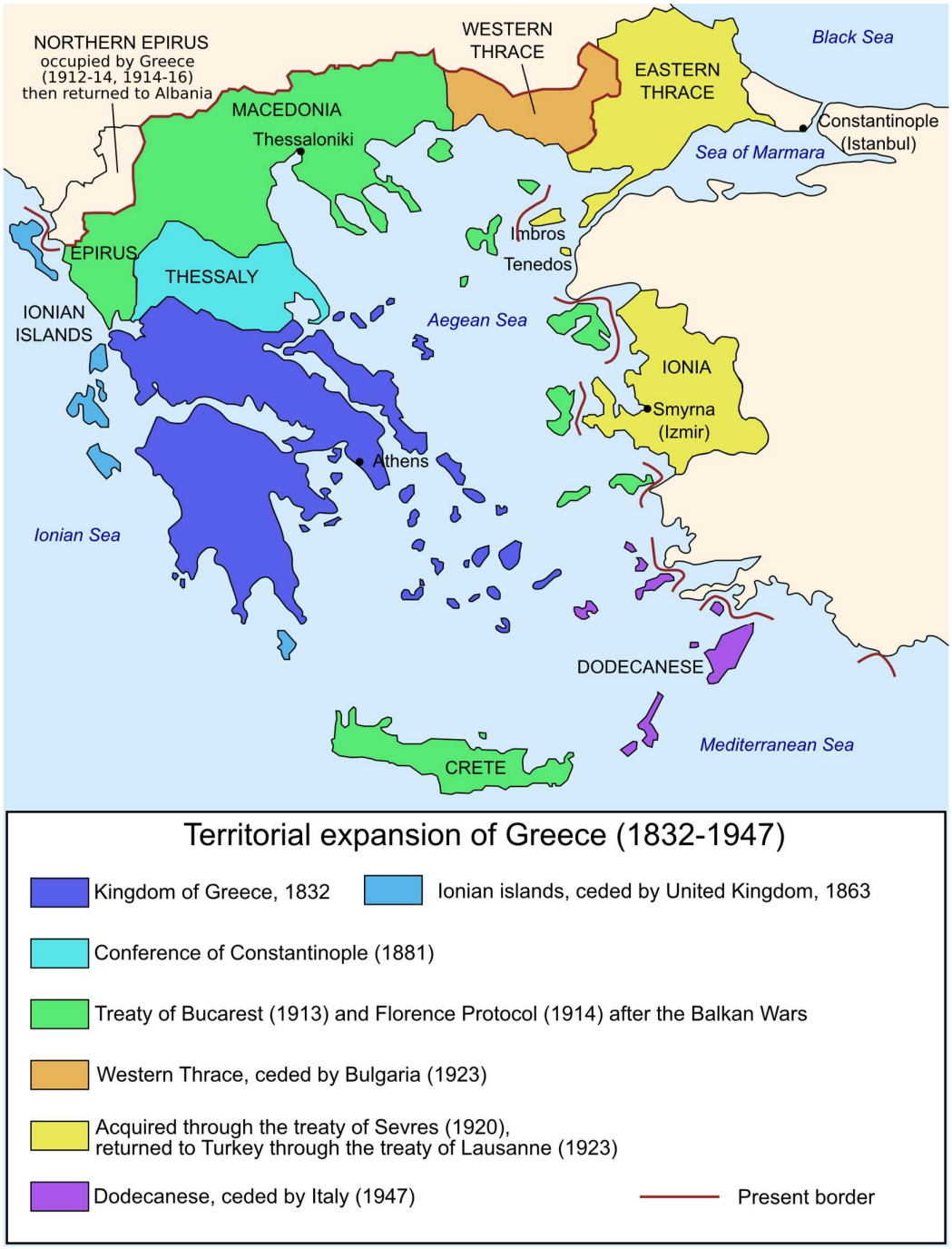
Source: ELSTAT, 1920 population census, 9.

Map 2: The Kingdom of Greece in 1921 and its territorial expansion till 1921.



Source: https://gallica.bnf.fr/ark:/12148/btv1b53209910p/f1#

Map 3: The territorial expansion of Greece 1832-1942.



Source: Wikipedia

SOURCES

1920 population census

The 1920 population census constitutes the 21st national census. It is the first to provide a more detailed picture of the population's demographic characteristics. In particular, it measured the de facto population of the country, i.e., the number of people present for whatever reason, at the time of the census, along with the legal population, meaning the number of Greek subjects, duly registered, present in Greece at the time of the census or found abroad. The population was also classified by gender, age, place of birth, nationality, municipality or commune of registration, marital status, education, profession, language maternal, certain physical infirmities, and religion.[2] This census did not refer to the active population, but rather to the real population with an occupation, and separately to the real population registered without one.

1928 population census

The 1928 population census followed the 1920 one. The aim was to record the refugee populations that arrived in Greece after the war of 1922 and the Treaty of Lausanne in 1923, which enforced the compulsory exchange of Greek Christians from Turkey and Muslim inhabitants of Greece. It was estimated that 1.221.849 refugees arrived in Greece, significantly augmenting the population. When it comes to the Muslim population that left Greece, they constituted a smaller part of the total population, with approximately 387.000 to 440.000 people leaving. The total population increased between 1920 and 1928, according to the 1928 census, and rose to 1.187.795 people.[3]

The 1928 census was also a fairly complete registration of the population's demographic characteristics, similar to the 1920 one, since it followed the same patterns. It is divided into four books, with the first addressing the real and legal population and the refugees, the second providing information on age, family, and

[2] Ministry of National Economy, General Statistical Service of Greece. "Population census of Greece in the 19 December 1920- General statistical results", Athens, 1928, p. 20, introduction. http://dlib.statistics.gr/Book/GRESYE_02_0101_00010.pdf

[3] Ministry of National Economy, General Statistical Service of Greece. "Statistical results of the 15th-16th May 1928 population census of Greece", v. III: Professions, issue 1, Athens, 1937, p. 8 introduction. http://dlib.statistics.gr/Book/GRESYE_02_0101_00020.pdf

education, the third on occupations, and the fourth on the place of birth, religion, and language.[4]

In the 1928 census, the active population included all individuals with an occupation, excluding those who do not make their living through market activities, such as those living off their inheritance and personal wealth, and those who depend on others' income, such as on the breadwinners'. That is to say, rentiers, pensioners, dependent women and children, and students were excluded from the active population. According to the census instructions, all women and children (over 10 years old) with an occupation were to be registered.[5] Specifically, it was mentioned:

> For women and children (over 10 years old) who work, their profession will also be declared. That is, if a woman works on her husband's land, she will also declare that she is a grain farmer, vine grower, etc., and the same applies to children. If she grazes sheep, she will declare that she is a livestock farmer; if she knits or weaves, she will also declare this, and so on. Only if she has no job that brings in income and is involved only in her household will she declare herself unemployed.[6]

However, despite this precise instruction, the percentage of women without a profession appeared suspiciously high.

1951 population census

The 1951 census registered the real and legal population in each municipality and region of the country, and their demographic characteristics, such as gender, age, place and country of birth, marital status, education, profession, native language, physical handicaps, and religion.

The active population consisted of the employed and unemployed people during the time of the census. The employed were considered a) the people who had any paid occupation or gainful activity the week that the census was taken, b) the members of the family who were helping the head of the family or any other member with their occupation during that week, and c) anyone who had a paid occupation or gainful

[4] Ibid., p. 8 introduction.
[5] Ibid.
[6] Ibid.

activity but had temporarily stopped working during the census due to illness, leave, strike, etc. Additionally, the occupational activity must last for an average of one hour per day. Unemployed were defined as those who were not working that week but were seeking paid employment. Finally, women working in the household and students were included in the non-active population if they did not have any other paid occupation, while also the pensioners, rentiers, and prisoners (even if they worked in prison for remuneration).[7]

1991, 2001 population censuses

Both the 1991 and 2001 censuses provide a good picture of the real and active population, especially in comparison with the previous censuses used for this database, and they categorize the population using the same criteria, simplifying comparisons. They both include similar demographic characteristics to those of the previous censuses, but are more detailed.

In these censuses, the active population included all employed and unemployed people over 10 years of age. More specifically, as employed were considered the people who a) were aged 10 years and over, and they work for profit or for pay, or as unpaid assistants in a family business or agricultural holding, provided that they work there every day for at least one-third of the regular working hours or more than 90 days a year, and, therefore, in the week prior to the census, worked more than 15 hours, b) people who responded that they work, but in the week before the census worked less than 15 hours or not at all (due to seasonality, weather conditions, leave, or illness). All people aged 10 or older who stated they are seeking work and taking steps in that direction were registered as "Unemployed". These were either individuals who had lost their jobs for any reason or people seeking employment for the first time. Both the 1991 and the 2001 censuses had a separate category for the latter called "Young".[8]

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[7] National Statistical Service of Greece (NSSG). "Population census of 7th April 1951 results", v. III: Tables of financial characteristics of population, Athens, 1958, p. 3-4, introduction. http://dlib.statistics.gr/Book/GRESYE_02_0101_00030.pdf

[8] National Statistical Service of Greece (NSSG). "Population- housing census results of 17th March 1991", v. III: Financial characteristics of population, Athens,1998, p. 16-18, introduction. http://dlib.statistics.gr/Book/GRESYE_02_0101_00082.pdf

METHODOLOGY

1920 dataset

For the 1920 dataset, I used data from the 1928 census, since it included tables for 1920, to compare them with the 1928 data. I considered the categorization of the information in those tables to be more detailed than that in the 1920 publication.

Firstly, I transcribed information about the occupations of the population, categorized by gender, economic branches, and social status, as appeared in Table 17[9]. The economic branches included in this table were: "Agriculture, Livestock farming-hunting, Fishery, Mining-Industry-Transports and communications, Credit, change and mediation- Commerce, Personal Services- Liberal Professions- Public Services, and No declared profession or imprecisely declared". The social statuses were: "Independents, Employees, and Workers"[10].

Secondly, I used Table 23[11], which divided the active population by gender, economic branches, and age. I wanted to combine the data from the two tables to determine the active population by gender, age, occupation type, and social status. To do so, I calculated the shares of social status and the shares of the different age groups per economic sector for males and females. Then I multiplied the shares of one age group (e.g., 10-19) by the shares of each social status (self-employed, employees, workers) and by the actual number of people (male or female over 10 years old) in each economic sector. An example of my methodology is illustrated in Table I, which refers to the male population aged 10 or older. In the first row, the actual numbers of people in each economic sector are shown; the second row shows the shares of 10- to 19-year-old males in each economic sector; and the third, fourth, and fifth rows show the shares of each social status in each economic sector. I multiplied the first row by the second and third, then the first by the second and fourth, and the first by the second and fifth. The figures resulting from these calculations are estimates of the number of working people in each age category, social status, and economic sector (see Table I). I followed the

[9] ELSTAT, population census 1928, Volume III, Issue I: Professions, Table 17, 70-71.
[10] According to the population censuses, the key distinction between an employee and a worker is that the former is primarily engaged in mental or clerical work, while the latter mainly performs manual or physical labor: Ministry of National Economy. General Statistical Service of Greece, "September 1930 census of industrial and commercial enterprises' employees and workers and their wages in comparison to previous and recent data", Athens, 1940, p. 7, introduction.
http://dlib.statistics.gr/Book/GRESYE_02_1903_00001.pdf
[11] ELSTAT, population census 1928, Volume III, Issue I: Professions, Table 23, 97-98.

same calculations for all age categories for men, and then for working women. Finally, for the "Independents", I assigned the labor relation code 12a013, since it was not clear how many were self-employed and how many were employers, and for the workers and employees, I assigned the code 14.

Table I. ESTIMATIONS

MALE	TOTAL	10-19	INDEPENDENTS	EMPLOYEES	WORKERS
Agriculture- Livestock farming- hunting- Fishery	817.539	19%	84%	1%	15%
Mining-Industry-Transports and communications	319.460	21%	25%	6%	69%
Credit, change, and mediation- Commerce	146.830	13%	66%	16%	17%
Personal Services- Liberal Professions- Public Services	117.328	14%	14%	64%	22%
No declared profession or vaguely declared	122.248	33%	1%	7%	92%

	12a013 INDEPENDENTS	14 EMPLOYEES	14 WORKERS
	130.479	1.553	23.300
	16.772	4.025	46.290
=	12.677	3.103	3.308
	2.300	10.513	3.614
	403	2.824	37.114

However, after taking a closer look at the labor force participation figures deriving directly for calculating the census data, it is evident that women's labor is significantly underrepresented. Only the 6th % of the female population appears engaging in the market economy.

Table II.

	MALE	FEMALE
INDEPENDENTS	35%	2%
EMPLOYEES	5%	0%
WORKERS	20%	3%
TOTAL	61%	6%

In the Greek bibliography, it has been frequently argued that the high figures of non-working women the censuses indicate are misleading and are attributed to under-

registration of women's work.[12] It was a common occurrence during the census-taking process that women's employment in the family was not recorded, with only the head of the family registered as the working person.[13] Additionally, the cottage industry was not included in the census, which traditionally employed a significantly high number of women.[14] Also, characteristic of the Greek industry during the interwar period was the abundance of small manual workshops and crafts with few workers or small family businesses.[15] Since this phenomenon of small-scale family shops is so frequently described in Greek literature, it is expected that some working women would hide among these family enterprises, assisting their husbands.

Therefore, I adopted a different method to estimate female labor force participation, aiming to overcome census under-registration of women in the labor force. To do so, I based my estimations on the research on women's work conducted by the anthropologist Susan Buck Sutton in Greece, and specifically in Nafplio, during the period 1920-1940. Nafplio during the interwar period was a small town, rather than the regional and national center it had been earlier. Sutton discovered that two-thirds of its married women population were working, and only one-third were staying at home. The distinction she finds between these women, who focused on family maintenance labor, and those who engaged in market-oriented labor was social class.[16]

Although Greece in 1920 was beginning to take preliminary steps toward industrialization, this process was confined to the country's two largest cities, Athens and Piraeus.[17] The rest of the country remained either rural, with the population being scattered in small villages rather than urban areas, or consisted of smaller provincial

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[12] Papastefanaki, Leda. 2016. "Labour in Economic and Social History: The Viewpoint of Gender in Greek Historiography." *Genesis : Rivista Della Società Italiana Delle Storiche : XV, 2, 2016*. Roma: Viella. https://doi.org/10.23744/1009, 66.

[13] Ibid., 66; Makris, Evaggelos. "O ikonomikos energos plithismos kai I apasholisis autou" [The economically active population and its employment] in *I statistiki kata ta 150 eti apo tis Paligenesias tis Ellados* [Statistical studies 1921-1971. Statistics during the first 150 years from the Rebearth of Greece], Athens 1972, 113- 212.

[14] Michalis Riginos, *Paragogikes domes kai ergatika imeromisthia stin Ellada 1909-1936* [Productive structures and workers' wages in Greece, 1909- 1936], Athens, Foundation of Research and Culture of the Commercial Bank of Greece, 1987, 136.

[15] Potamianos Nikos, *Oi noikokiraioi: Magazatores kai viotehnes stin Athina 1880-1925 [The householders: Shopkeepers and artisans in Athens 1880-1925]*, (Heraclion: Crete University Press, 2015), 10.

[16] Sutton, Susan Buck. 1985. "Women's Work in Nafplio, 1920-1940." *Ethnohistory* 32 (4): 343. https://doi.org/10.2307/481894.

[17]Leodidou Lila, *Polis tis Siopis, ergatikos epoikismos tis Athinas kai tou Peiraia, 1909-1940, [Cities of silence, workers' settling of Athens and Piraeus, 1909-1940]*, Cultural Technological Foundation ETVA, 1989, 62-64.

towns.[18] Only in larger cities were women slowly confining themselves to household labor, fitting the ideal of domesticity,[19] while in the countryside, most women worked alongside the entire family in small-scale farming operations.[20] Since Nafplio was a provincial town and most of Greece was still not fully urbanized, I considered it safe to aggregate her research results for the whole country.

First, I calculated two-thirds of the women registered in the "Without occupation" category. They accounted for 1.651.516 people, so the 2/3 were 1.101.011 females. This is the number of the extra women who would participate in the labor market but were excluded from it in the census. I added this number to the women already registered with an occupation. To determine how these extra working women would be allocated across the different economic sectors, I calculated the percentages of the registered women in each economic branch and then multiplied the number of extra working women by these shares.

Table III.

EXTRA W.W.			
Agriculture- Livestock farming- hunting- Fishery	1.101.011	32%	348.311
Mining-Industry-Transports and communications	1.101.011	18%	195.108
Credit, change, and mediation- Commerce	1.101.011	1%	12.981
Personal Services- Liberal Professions- Public Services	1.101.011	14%	149.350
No declared profession or imprecisely declared	1.101.011	36%	395.260

Then I added these figures to the registered women across the different branches, and I arrived at the new figures for women per economic sector.

Table IV.

NEW W.W.	Registered w.w.	Extra w.w.	NEW W.W
Agriculture- Livestock farming- hunting- Fishery	108.775	348.311	457.086
Mining-Industry-Transports and communications	60.931	195.108	256.039
Credit, change, and mediation- Commerce	4.054	12.981	17.035
Personal Services- Liberal Professions- Public Services	46.641	149.350	195.991
No declared profession or imprecisely declared	123.437	395.260	518.697
TOTAL	343.838	1.101.011	1.444.849

[18] Leodidou, *Cities of silence*, 58,59.
[19] Sutton, "Women's Work in Nafplio", 346.
[20] Ibid.

The next step was to calculate their age distribution and determine how many would have been considered 'Independents', 'Employees', or 'Workers'. Therefore, I calculated the shares of each age category within each economic sector from Table 23, as well as the shares of each 'Status in the profession' in each economic sector from Table 17 in the census. Then I multiplied the shares of each age category by the shares of each social status and by the figures of the 'New Working Women' in each economic sector. This is how I found the final distribution of these estimated working women across different sectors, ages, and work positions, and assigned the labor relations codes.

Table V.

INDEPENDENTS (12a013)	10- 19	20-29	30-39	40-49	50-59	over 60
	97.908	75.373	70.323	61.387	42.738	40.795
	17.001	11.225	5.285	3.687	2.253	1.516
	1.417	1.843	1.226	1.226	744	617
	4.204	2.940	941	647	588	500
	10.177	6.536	4.979	4.046	2.552	2.801
total	130.707	97.916	82.754	70.992	48.875	46.228
EMPLOYEES (14)						
	1.152	887	827	722	503	480
	3.188	2.105	991	691	422	284
	1.250	1.626	1.082	1.082	656	544
	15.975	11.171	3.575	2.458	2.234	1.899
	3.392	2.179	1.660	1.349	851	934
total	24.957	17.967	8.135	6.301	4.667	4.141
WORKERS (14)						
	16.126	12.414	11.583	10.111	7.039	6.719
	86.068	56.825	26.754	18.665	11.407	7.674
	739	961	640	640	388	322
	63.901	44.686	14.300	9.831	8.937	7.597
	156.045	100.212	76.352	62.036	39.131	42.948
total	322.879	215.099	129.627	101.283	66.902	65.259

The final female labor force participation percentage, following this method, rose to 57% of the total female population in 1920.

Table VI.

INDEPENDENTS	19%
EMPLOYEES	3%
WORKERS	36%
TOTAL	57%

The remaining non-working women over 10 years old totaled 550.505, and I assigned them the labor relation code 5. The non-working men were 399.251 and were assigned labor relation code 1. For children under 10 years old, I examined their literacy status and assumed that many of the 5-9-year-olds who were illiterate would most likely work alongside their family members. Therefore, I calculated ½ of the illiterate male and female children 5-9 years old as a minimum, and I assigned labor relation 12b. For the rest of the children under 10 years old, I assigned labor relation 1.[21]

1928 CENSUS

For the 1928 population census, I followed a similar methodology to that of the 1920 census. I used Table 23,[22] which presents the real population over 10 distributed by economic sector and age, and Table 17,[23] which presents the productive population by social status and economic sector. I assigned labor relations codes based on different social statuses. More specifically, the social statuses included were: “Employers”, “Self-Employed”, “Members of the family’s employer”, “Employees”, and “Workers”. I assigned code 13 to employers, 12a to the self-employed, 12b to the employer's family members, and 14 to employees and workers. This could be done immediately just by taking the data from Table 17. However, to create a more complete dataset that includes information on the population's age distribution, I combined the data from these two tables.

To do so, I multiplied the total number of people working in a specific occupation category (male and female) by the percentage of people of a particular age category (eg, 10-19) and occupation category, and by the percentage of the people with a specific position in the profession and occupation category. From this estimated count of people by age, economic category, and social status, I got the following results.

[21] See Table XV.
[22] ELSTAT, population census 1928, Volume III, Issue I, Table 23, 96-97.
[23] ELSTAT, population census 1928, Volume III, Issue I, Table 17, 70-73.

Table VII.

LABOR RELATIONS TABLE		MALE		FEMALE	
EMPLOYERS	13	469.272	15%	107.048	3%
SELF-EMPLOYED	12a	500.823	16%	106.783	3%
Members of the family's employer	12b	287.338	9%	272.493	9%
EMPLOYEES	14	148.596	5%	27.924	1%
WORKERS	14	556.592	18%	255.928	8%
TOTAL		1.962.621	64%	770.176	25%

Even though the census argues that all women with a gainful occupation were to be registered, the female labor force participation appears to be only 25% of the total female population. In order to cross-check these results and determine whether a significant portion of the female labor force had been excluded from the census, I used a different method.

I again based my estimations on Susan Buck Sutton's anthropological research, since it also covers the year 1928.[24] Therefore, I again departed from the assumption that at least two-thirds of the women aged 10 years old or older who were registered as having no occupation were working. This means that from the 1.675.858 women appearing in the 1928 census in the category "Without an occupation", I calculated 2/3 of them: 1.117.239 for the dataset. Following, I wanted to estimate how these new working women were distributed across the different economic sectors, age categories, and economic positions, so that I could identify their labor relations codes and their occupations, as well as their ages. Therefore, I calculated the shares of women across the different economic sectors from Table 23,[25] and I multiplied them by 1.117.239. In this way, I found the number of these women, who were initially registered without a profession, in each economic sector, as appears in the Table below.

[24] Sutton, "Women's Work in Nafplio", 352.
[25] ELSTAT, population census 1928, Volume III, Issue I, Table 23, 96-97.

Table VIII.

	EXTRA WORKING WOMEN OVER 10	WORKING Women OVER 10 from the 1928 census	TOTAL
Agriculture	627.942	434.623	1.062.565
Livestock farming- hunting	47.709	33.021	80.730
Fishery	59	41	100
Mining	526	364	890
Industry	144.064	99.712	243.776
Transport and communications	1.321	914	2.235
Credit, change, and mediation	2.563	1.774	4.337
Commerce	7.163	4.958	12.121
Personal Services	54.002	37.377	91.379
Liberal Profesions	26.339	18.230	44.569
Public Services	3.498	2.421	5.919
No declared profession or imprecisely declared	202.053	139.849	341.902
TOTALS	1.117.239	773.284	1.890.523

In the first row, I show the estimated numbers of the extra working women, which I then add to those who appeared in the census (second row) to get my final figures. These are the ones I used for the dataset. To be able to provide their age and the labor relation code in the dataset, I multiplied the new number of working women by the shares of women in each economic sector within an age group (eg, in the age group 10-19 years old, 22% of women were occupied in Agriculture, 46% in Industry etc) and by the shares of each 'position in the profession' within a specific economic sector (Employers, Self-employed, Members of the family's employer, Employees, and Workers). The new figures for women's labor relations in 1928 are:

Table IX.

Employers	261.711	8%
Self-employed	261.063	8%
Members of the family's employer	666.190	21%
Employees	68.268	2%
Workers	625.692	20%
TOTAL	1.882.924	60%

With these new estimations, the Female Labor force participation rose to 60 % from the 22% that the census had initially registered.

The number of non-working women over 10 years was 558.619, and I assigned them the labor relation code 5. The non-working men accounted for 393.354, and I assigned lab.rel. 1. For the children under 10 years old, I followed the same method as in the 1920 census. That is to say, I estimated that half of the children aged 5-9 who had been registered as illiterate would be assisting their families, and I assigned them labor relation 12b. For the rest of the population under 10 years old, I assigned labor relation 1.

1951 CENSUS

For the 1950 cross-section, I also used information from the 1951 population census. To discover the different labor relations from this census, I used the census's Table 3[26] with information about the distribution of the active population into the different economic sectors and their status in the profession ('Employers', 'Self-employed', 'Employees', 'Working family members of the employer', and 'Not declared'). I assigned the labor relations codes as follows:

Table X.

Employers	13
Self-employed	12a
Wage workers	14
Assisting family members	12b

To obtain more complete information about the population included in the dataset, I combined this information with Table 6[27], which provided details on the distribution of the active male and female populations by social status and age. Firstly, I calculated the shares of the people in each age category and social status from Table 6. Then, I multiplied each share by the actual number of people in each social status and economic branch, as shown in the census Table 3. For instance, the number of male employers aged 20-24 was 3.164 in Table 6 of the census. I divided 3.164 by 72.644, which is the total of the male employers, to find out that they accounted 4%. Then I multiplied this 4% by 16.471, the number of male employers in the economic branch 'Agriculture, livestock farming- hunting, fishery', to get the number 717. Therefore, 717 men aged 20 to 24 years held the position of employer in Agriculture, livestock farming/hunting,

[26] ELSTAT, population census 1951, Volume III, Table 3, 125 (134).
[27] ELSTAT, population census 1951, Volume III, Table 5, 162 (172).

and fishery. I followed this process for all economic branches, social statuses, age categories, and genders. The dataset I provided for the male population in 1951 was derived from the calculations above. The figures for both the male and female populations in 1951, along with the labor relation codes, as derived directly from the census data, are shown in the table below.

Table XI.

1951		MALE	FEMALE	MLFP	FLFP
EMPLOYERS	13	72.644	4.483	2%	0%
SELF-EMPLOYED	12a	982.351	84.856	26%	2%
WAGE WORKERS	14	819.825	236.284	22%	6%
ASSISTING FAMILY MEMBERS	12b	309.002	152.238	8%	4%
NOT DECLARED	12a012b013014	145.079	32.719	4%	1%
TOTAL WORKING		2.328.901	510.580	63%	13%

The female labor force participation from the census appears to reach only 13% of the total female population. This percentage is low, and therefore, I proceeded to re-estimate it. I followed the same method as in 1928. Basing my estimations on Sutton's work seemed reasonable, even though her research referred to an earlier period (1920-1940) compared to the 1951 census. Since the research was conducted in a provincial town, where economic transformations occurred more gradually and less dramatically than in Athens, it is reasonable to assume that societal values and behaviors would have undergone fewer changes over the 10 years.[28]

The 1951 census, unlike the 1920-28 ones, also distinguished the population by urban, semi-urban, and rural areas. This allowed me to make more precise assumptions regarding the percentages of the extra women who would have been working. In particular, for semi-urban areas, I again assumed that 2/3 of the women registered as "Without a profession" would actually have been working. For the urban female population, I made a different assumption: one-third of women with no profession have a paid occupation. I chose a lower percentage as a starting point, taking into

[28] Sutton, "Women's Work in Nafplio", 346.

consideration that Greece had urbanized significantly during that period, with waves of migration from villages to cities, especially to the capital,[29] along with Shutton's research on women who had migrated from villages to Athens, showing that many of them changed their working patterns after migrating to the city. Even though village women generally worked alongside their husbands after marriage, migrant women were often involved in wage work before marriage and afterwards devoted themselves to household labor.[30] Therefore, I chose to add a smaller proportion of women living in the city to the extra female labor force, but this should be taken as a minimum. For the rural population, on the other hand, I assumed that ¾ of women would work, since women in the countryside would always need to perform agricultural activities and thus work in the fields.[31]

The 1951 census provided more detailed data on the registered non-active population than the 1920 and 1928 censuses. More specifically, in the census Table 13[32] the number of women over 10 years old without an occupation reached 2.715.492, while the total registered non-active female population rose to 3.400.573. This population was then divided in the census into categories such as dependent on their families, on social institutions, or having other types of income. Among the female population that appeared to depend on their families, there were two categories of women who conducted household labor, one with those who had a servant and the other with those who had no servants at all. I counted the number of women involved in household labor who had no servants and, depending on the society they lived in (urban, semi-urban, or rural), I calculated 1/3, 2/3, and 3/4 accordingly. The following table shows the economically inactive women engaged in household labor without a servant, and the next row shows my calculation of how many of them would actually be involved in the market economy but were omitted from the census.

[29] Leodidou, *Cities of silence,* 156-161, 179.
[30] Susan Buck Sutton, "Family and Work: New Patterns for Village Women in Athens," Journal of Modern Greek Studies 4, no. 1 (May 1, 1986): 33–49, https://doi.org/10.1353/mgs.2010.0064, 33.
[31] Sutton, "Women's Work in Nafplio", 345.- Sutton, "Family and Work: New Patterns for Village Women in Athens." 33.
[32] ELSTAT, population census 1951, Volume III, Table 13, 350 (358).

Table XII.

HOUSEWIVES WITHOUT A SERVANT		
	Total	Extra working women
URBAN	776.337	258.779
SEMI-URBAN	344.986	229.991
AGRICULTURAL	1.112.557	834.418
TOTAL	2.233.880	1.323.187

Women with a market-oriented occupation rose from 510.580, as registered in the census, to 1.833.767. The FLFP rose from the previous 13% to 47%.

For the non-working population, I used again the data from Table 13 of the census on the inactive population.[33] The categories that the census had for the non-active population were: "Dependents on their families, Supported by Social Institutions, and Supported by other resources, such as pensioners, and rentiers". For the male population in all of those categories, I assigned labor relation 1. The category "Dependents of their families" had the subcategories: "Housewives without a servant, Housewives with a servant, Students, Not able to work due to young age, Not able to work due to old age, and Not able to work due to handicap." For the "Housewives without a servant," I subtracted the ones that I included before in the working population, and for the rest, I assigned labor relation 5. I also assigned code 5 to "Housewives with a servant". I assigned labor relation 1 to the rest of these women.

For children under 10 years old, I used data from Table 9 of the census[34] regarding the population's demographic characteristics (See table below).

Table XIII.

1951	Illiterate		Literate		Not declared	
	M	F	M	F	M	F
TOTAL GR	507.479	485.172	209.885	191.525	8.822	8.384
URBAN GR	154.204	148.145	58.990	55.586	3.163	3.019
SEMI-U GR	77.862	73.371	33.292	30.766	1.507	1.426
RURAL GR	275.413	263.656	117.603	105.173	4.152	3.939

Source: ELSTAT, population census 1951, Volume II, Table 9, 294-300 (300-306).

[33] Ibid.
[34] ELSTAT, population census 1951, Volume II, Table 9, 294-300 (300-306).

I assumed, as a minimum, that half of the illiterate children living in semi-urban and rural areas would be working but would have been excluded from the census. I did not include illiterate children from urban areas, since I could more confidently assume that children from families involved in agricultural activities would assist with the agricultural labor. I assigned labor relation 12b to these children, and for the remaining ones, I assigned labor relation 1.

1991-2001 CENSUSES

The 1991 and 2001 censuses included the most complete information compared to the censuses used above. Under-registration of women's participation in the labor market must exist, but I assume it will be less significant than in previous years. Firstly, because of the progressive shift in social norms and opinions surrounding women in the public sphere, of women performing different occupations, and, by extension, the census-taking process.[35] However, I did not re-estimate women's labor force participation for these two datasets, since alternative sources should be used to reveal more accurate figures. I assigned the labor relations codes as follows:

Table XIV.

EMPLOYERS	13
SELF EMPLOYED	12a
WAGE WORKERS	14
ASSISTING UNPAID HOUSEHOLD MEMBERS	12b
YOUNG	3
NON-WORKING (over 10 years)	1
HOUSEWIVES	5
CHILDREN (under 10)	1

Since I used different methods across the cross-sections, a precise comparison of changes in labor force participation and labor relations' development throughout the 20th and early 21st centuries is difficult. It is possible to follow some general trends. Additionally, although the figures on women's work constitute a point of departure for reconstructing female labor force participation, alternative sources and further research

[35] Nederveen Meerkerk, Elise van, and Richard Paping. 2014. "Beyond the Census. Reconstructing Dutch Women's Labour Market Participation in Agriculture in the Netherlands, ca. 1830–1910." *The History of the Family* 19 (4): 447–68. doi:10.1080/1081602X.2014.955515. 450.

are essential to reach more accurate conclusions. However, the final labor relations that appeared across all cross-sections are presented below.

Table XV.

	Lab.rel	MALE	FEMALE (recalculated)	% MALE	% FEMALE (recalculated)
1920					
Independents	12a013	881.761	477.472	35%	19%
Employees	14	134.860	66.168	5%	3%
Workers	14	506.784	901.048	20%	36%
Total working (10+ years)		1.523.405	1.444.688	61%	57%
Non-working (10+ years)	1(M), 5(F)	399.251	550.505	16%	22%
Working children (under 10)	12b	88.605	96.909	4%	4%
Non-working children (under 10)	1	486.609	431.657	19%	17%
Rest of women	-1	-	161	-	0,01%
Rest of men	-1	0	-	0%	-
Real population men		2.497.870			
Real population women			2.523.920		
Real population total		5.021.790			
1928		MALE	FEMALE		
Employers	13	469.272	261.711	15%	8%
Self-employed	12a	500.823	261.063	16%	8%
Members of the employer's family	12b	287.338	666.190	9%	21%
Employees	14	148.596	68.268	5%	2%
Workers	14	556.592	625.692	18%	20%
Total working (10+ years)		1.962.621	1.882.924	64%	60%
Non-working (10+ years)	1(M), 5(F)	393.354	558.619	13%	18%
Working children (under 10)	12b	73.400	79.278	2%	3%
Non-working children (under 10)	1	637.257	600.029	21%	19%
Rest of women	-1	-	7.599		0,2%
Rest of men	-1	9.603	-	0,3%	

Real population men		3.076.235			
Real population women			3.128.449		
Real population total		6.204.684			
1951		MALE	FEMALE		
Employers	13	72.644	16.101	2%	0,4%
Self-employed	12a	982.351	304.764	26%	8%
Wage workers	14	819.825	848.623	22%	22%
Assisting family members	12b	309.002	546.769	8%	14%
Not declared	12a012b013014	145.079	117.512	4%	3%
Total working (10+)		2.328.901	1.833.767	63%	47%
Non-working (10+)	1	666.561	477.673	18%	12%
Housewives	5	0	914.632	0%	23%
Working children (under 10)	12b	314.344	205.199	8%	5%
Non-working children (under 10)	1	411.842	479.882	11%	12%
Rest of women	-1	-	0	-	0%
Rest of men	-1	0	-	0%	-
Real population men		3.721.648			
Real population women			3.911.153		
Real population total		7.632.801			
1991		MALE	FEMALE		
Employers	13	214.947	47.993	4%	1%
Self employed	12a	889.355	205.849	18%	4%
Wage workers	14	1.299.578	712.899	26%	14%
Assisting unpaid household members	12b	94.549	136.725	2%	3%
Young	3	81.320	87.096	2%	2%
No declared	-1	70.921	44.925	1%	1%
Total working (10+)		2.650.670	1.235.487	52%	24%
Non-working (10+)	1	1.672.784	1.576.634	33%	30%
Housewives (and "other women")	5	0	1.798.587	0%	35%
Children (under 10)	1	627.221	593.200	12%	11%
Rest of women	-1	-	584	-	0,01%
Rest of men	-1	104.733	-	2%	-
Real population men		5.055.408			

Real population women			5.204.492		
Real population total		10.259.900			
2001		MALE	FEMALE		
Employers	13	370.764	131.453	7%	2%
Self employed	12a	568.189	181.445	10%	3%
Wage workers	14	1.752.810	1.142.128	32%	21%
Assisting unpaid household members	12b	47.735	173.549	1%	3%
Young	3	137.378	109.048	3%	2%
Total working		2.876.876	1.737.623	53%	31%
Non-working (10+)	1	1.984.760	1.675.872	37%	30%
Housewives (and "other women")	5	0	1.584.462	0%	29%
Children (under 10)	1	551.790	522.714	10%	9%
Rest of women	-1	-	0	-	0%
Rest of men	-1	0	-	0%	-
Legal[36] population men		5.413.426			
Legal population women			5.520.671		
Legal population total		10.934.097			

[36] It is important to highlight that for the year 2001, the total male and female population in the census publication does not refer to the real population, as in the rest of the datasets, but to the legal population. The real population was not included in the publication used for this dataset.

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